MILITARY STANDARD

MIL-STD-1320-106A (NAVY)

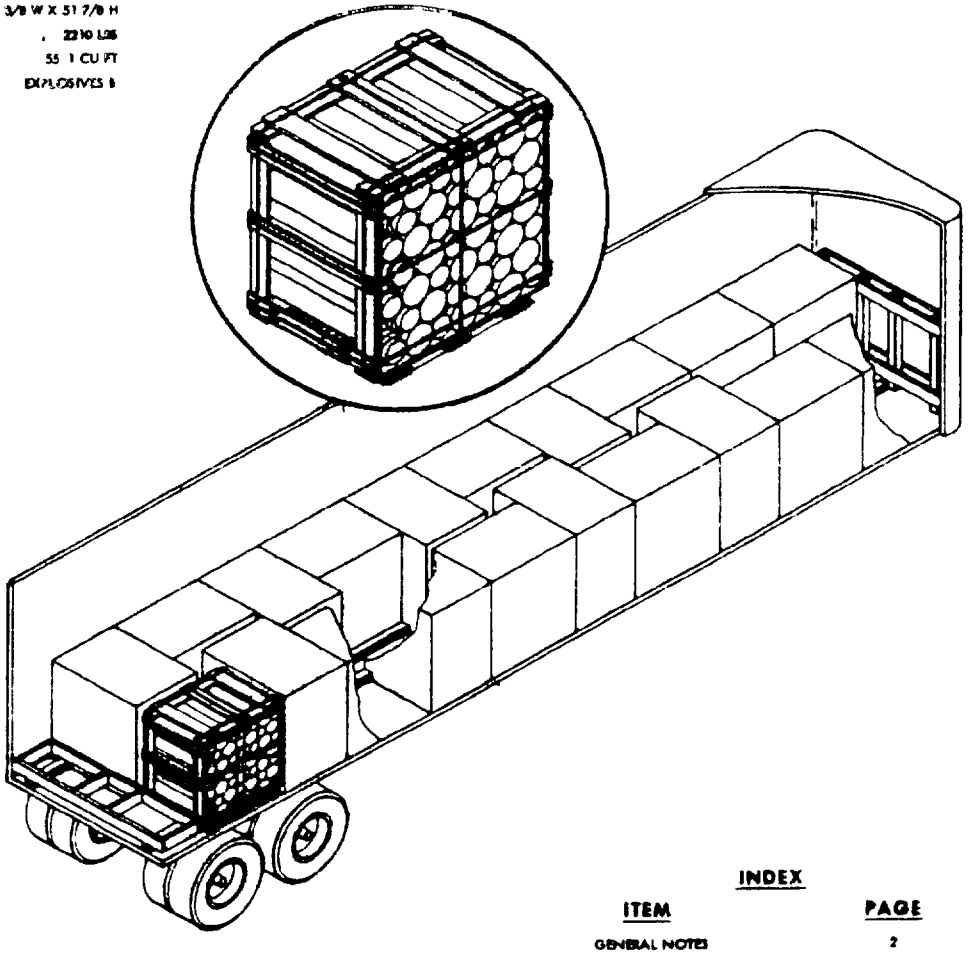
25 MARCH 1976

SUPERSEDING
MIL-STD-1320-106 (NAVY)
10 SEPTEMBER 1974

TRUCKLOADING
CHARGE, PROPELLING, 5"/54
DOMESTIC UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-53/753
DIMENSIONS	50 1/2 L X 36 3/8 W X 51 7/8 H
WEIGHT (APPROX)	2210 LBS
CUBE	55 1 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES B



NOTES

1 UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES

2 FOR CROSS-REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY)

INDEX

ITEM	PAGE
GENERAL NOTES	2
LOAD PATTERNS	3
TYPICAL LOADS	4, 5, 6 & 7

FSC 8140

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M S Gray NWHL 3/22/76
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J C Kelly 3/23/76
SIGNATURE SEA SYSCOM, BY DIRECTION DATE

ORIGINATOR L L Clarington 3/22/76
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE PROPELLING CHARGE 5"/54 IN TANK MK 14 THE TANKS SHALL BE PALLETIZED IN ACCORDANCE WITH WR-53/753
2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 42 44 AND 45 FT TRAILERS THE TRAILERS MAY HAVE FLOORS THAT ARE ALL WOOD ALL METAL OR METAL FLOORS WITH WOOD NAILING STRIPS THE TRAILER S TANDEM AXLES SHALL BE LOCATED IN THE "WESTERN POSITION (AT THE EXTREME REAR OF THE TRAILER)
3. THE WIDTH OF THE SLEEPERS PIECE NO 8 ON PAGE 3 SHALL BE SUFFICIENT TO FILL THE VOID BETWEEN THE PALLET RUNNERS A MAXIMUM OF ONE INCH SLACK IS PERMITTED
4. THE WIDTH OF THE FILLER ASSEMBLIES DETAILS F G J & K SHALL FILL THE VOID BETWEEN UNIT LOADS OR UNIT LOAD AND SIDE WALL AS APPROPIATE A MAXIMUM OF ONE INCH IS PERMITTED
5. THE TYPE OF REAR BLOCKING TO BE USED IS DEPENDENT UPON THE DISTANCE BETWEEN THE TRAILER DOORS WHEN CLOSED AND THE REAR OF THE LADING

   DISTANCE LESS THAN 12 IN

   INSTALL REAR BLOCKING ASSEMBLY DETAIL D AS SHOWN ON PAGE 5

   CAUTION

   REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED DO NOT USE TRAILERS WITH ROLL UP DOORS

   DISTANCE 12 TO 36 IN

   INSTALL REAR BLOCKING ADDEMBLY DETAIL B AS SHOWN ON PAGE 4

   CAUTION

   REAR BLOCKING MUST BEAR AGAINST LADING AND TRAILER DOORS WHEN DOORS ARE CLOSED DO NOT USE TRAILERS WITH ROLL UP DOORS

   DISTANCE GREATER THAN 36 IN

   INSTALL REAR BLOCKING DETAIL H AS SHOWN ON PAGE 6 DO NOT USE TRAILERS WITH ALL METAL FLOORS TRAILERS WITH ROLL UP DOORS MAY BE USED
6. THE MAXIMUM GROSS WIEGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR TRAILER WILL NOT EXCEED THESE LIMITATIONS
7. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA CLOSE AND SEAL TRAILER DOORS AND ATTACH "EXPLOSIVES B" PLACARD TO BOTH SIDES AND BACK OF TRAILER AND FRONT OF TRACTOR
8. APPLICABLE MATERIAL SPECIFICATIONS DUNNAGE LUMBER, MM-L-751 NAILS FF-N-105 TYPE II, STYLE 10
9. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

FTL & LTL ALL LENGTH TRAILERS (VAN)

1. A FULL TRUCKLOAD CONSISTS OF
   - A 20 UNIT LOADS IN A LIGHT WEIGHT TRACTOR-TRAILER (TARE VEHICLE WEIGHT APPROXIMATELY 25 500 LBS)
   - B 19 UNIT LOADS IN A STANDARD WEIGHT TRACTOR-TRAILER (TARE VEHICLE WEIGHT APPROXIMATELY 27 500 LBS)

   NOTE

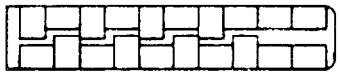
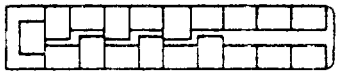
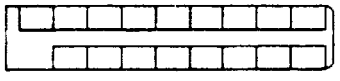
   20 UNIT LOAD MAY BE LOADED INTO A STANDARD WEIGHT TRAILER WHEN ROUTING IS THROUGH STATES PERMITTING 34 000 LBS OR MORE ON TANDEM AXLES AND A GROSS VEHICLE WEIGHT OF 80 000 LBS
2. WHEN LOADING A TRAILER
   - A SELECT THE CORRECT TABLE (TABLE I II OR III) FOR THE LENGTH OF TRAILER BEING LOADED
   - B FROM THE SELECTED TABLE, DETERMINE THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE LOADED
   - C LOAD THE TRAILER USING THE SELECTED LOAD PATTERN BLOCK AND BRACE USING THE PRINCIPLES OF THE SAMPLE LOADS SHOWN ON PAGES 4 5 6 AND 7
3. WHEN TRAILERS HAVE NAILABLE FLOORS SLEEPERS MAY BE USED TO CONTROL LATERAL MOTION AS SHOWN ON PAGES 4 AND 6
4. WHEN TRAILERS HAVE NON-NAILABLE FLOORS FILLER ASSEMBLIES SHALL BE USED TO CONTROL LATERAL MOTION AS SHOWN ON PAGES 5 AND 7

FTL-TRUCKLOAD DATA

TYPE OF TRAILER	NO OF UNIT LOADS	TRUCKLOAD WT (APPROX)
LIGHT WT	20	44 200
STD WT	19	42, 180

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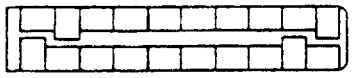
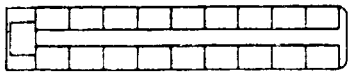
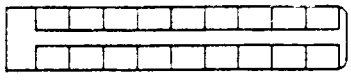
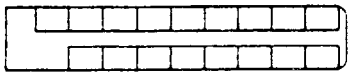
TABLE I
40 FT TRAILERS

NUMBER OF UNIT LOADS	LOAD PATTERN → FWD
* 20	
** 19	
18	
17	
16 OR LESS	CONTINUE USING DOUBLE ROW PATTERN

* SEE FTL NOTE 1

** FOR REAR BLOCKING PROCEDURE SEE PAGE 7

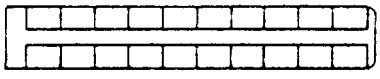
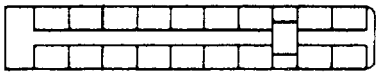
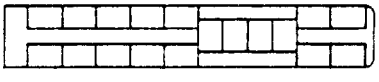
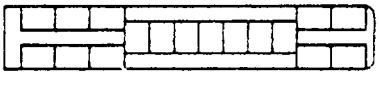
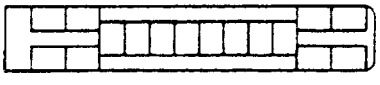
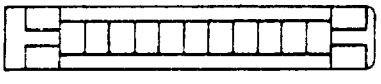
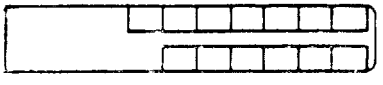
TABLE II
42 FT TRAILERS

NUMBER OF UNIT LOADS	LOAD PATTERN → FWD
* 20	
** 19	
18	
17	
16 OR LESS	CONTINUE USING DOUBLE ROW PATTERN

* SEE FTL NOTE 1

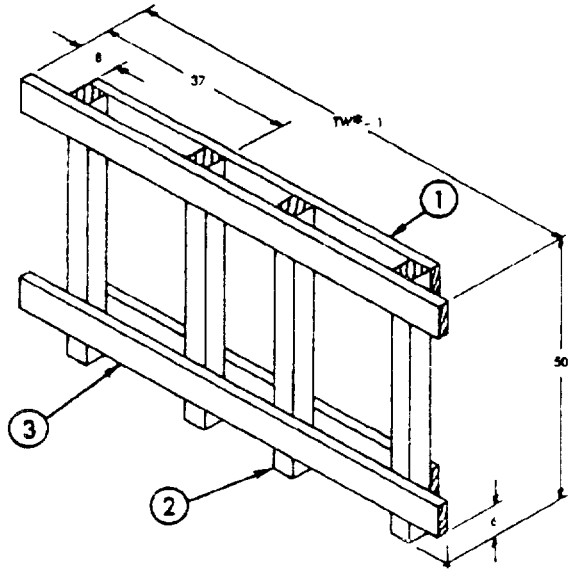
** FOR REAR BLOCKING PROCEDURE SEE PAGE 7

TABLE III
44/45 FT TRAILERS

NUMBER OF UNIT LOADS	LOAD PATTERN → FWD
* 20	
19	
18	
17	
16	
15	
14	
13	
12 OR LESS	CONTINUE USING DOUBLE ROW PATTERN

* SEE FTL NOTE 1

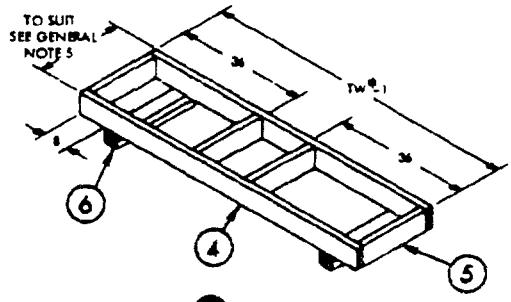
* TW = TRAILER WIDTH



DETAIL A

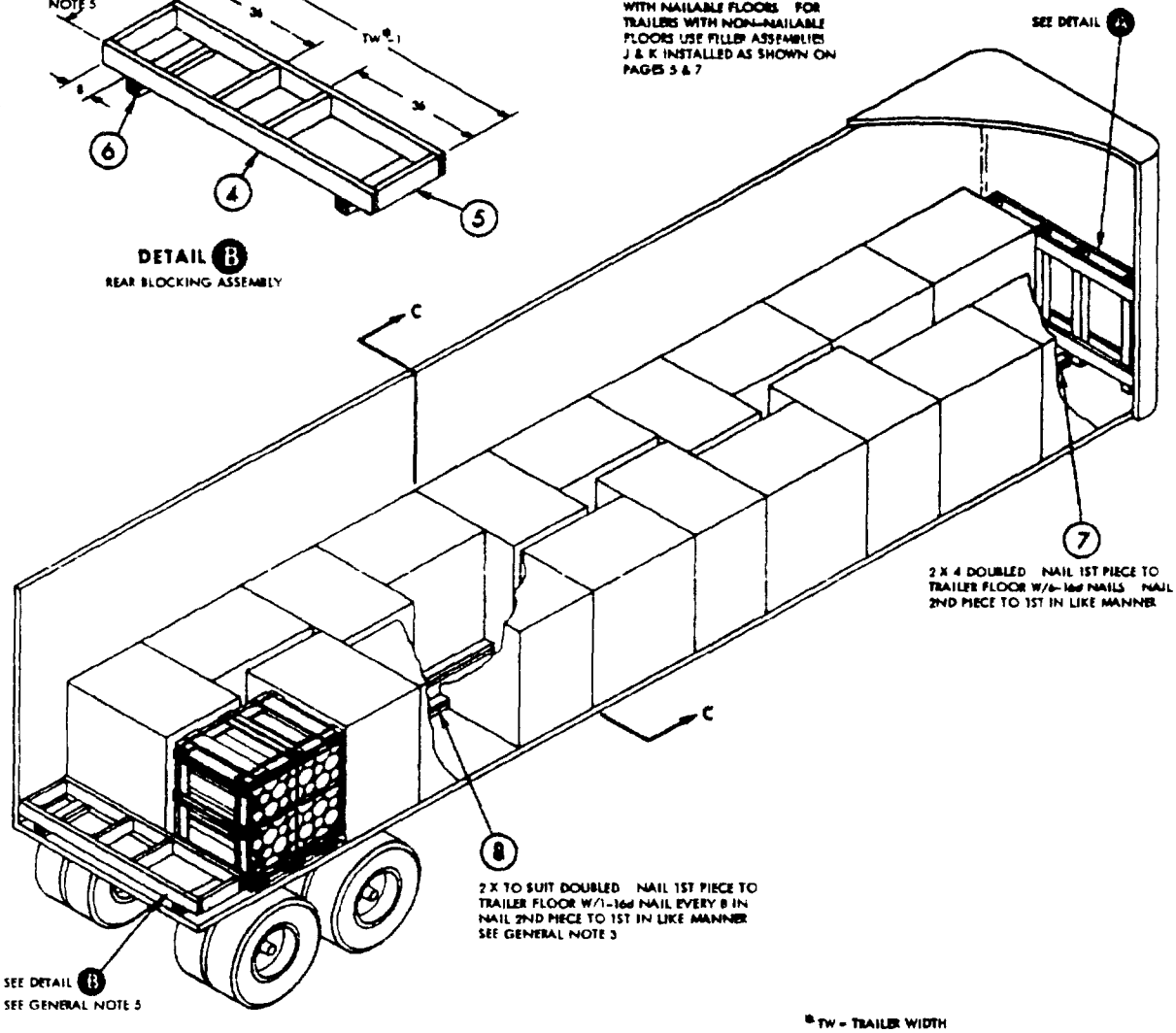
FRONT BULKHEAD
(1 REQUIRED)

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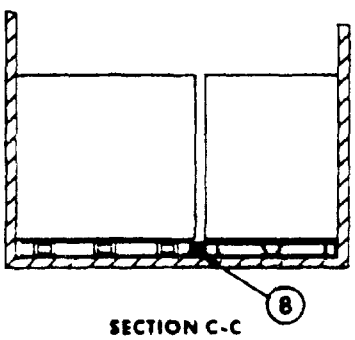


DETAIL B
REAR BLOCKING ASSEMBLY

NOTE
THIS TRUCKLOAD IS FOR TRAILERS WITH NAILABLE FLOORS FOR TRAILERS WITH NON-NAILABLE FLOORS USE FILLER ASSEMBLIES J & K INSTALLED AS SHOWN ON PAGES 5 & 7



* TW = TRAILER WIDTH

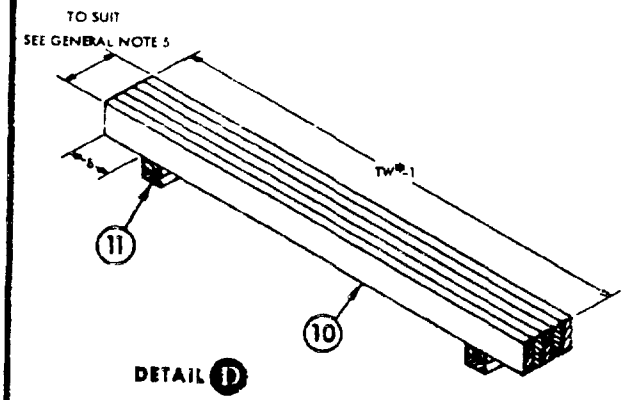


SECTION C-C

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
8	SLEEPER	2 X WIDTH TO SUIT X 48	16	SEE FIELD NOTE AND GENERAL NOTE 3		
7	SLEEPER	2 X 4 X 48	8	SEE FIELD NOTE		
6	SUPPORT	2 X 4 TO SUIT	4	4	3/JOINT	16d
				8	3/PIECE	16d
5	STRUT	2 X 6 TO SUIT	4	SEE 4	-	-
4	CROSSMEMBER	2 X 6 X TW*-1-	2	5	3/JOINT	16d
3	AFT CROSSMEMBER	2 X 6 X TW*-1	2	2	3/JOINT	12d
2	UPRIGHT	4 X 4 X 50	4	SEE 2 & 3	-	-
1	FWD CROSSMEMBER	2 X 6 TO SUIT	2	2	3/JOINT	12d

LIST OF MATERIALS & NAILING DATA

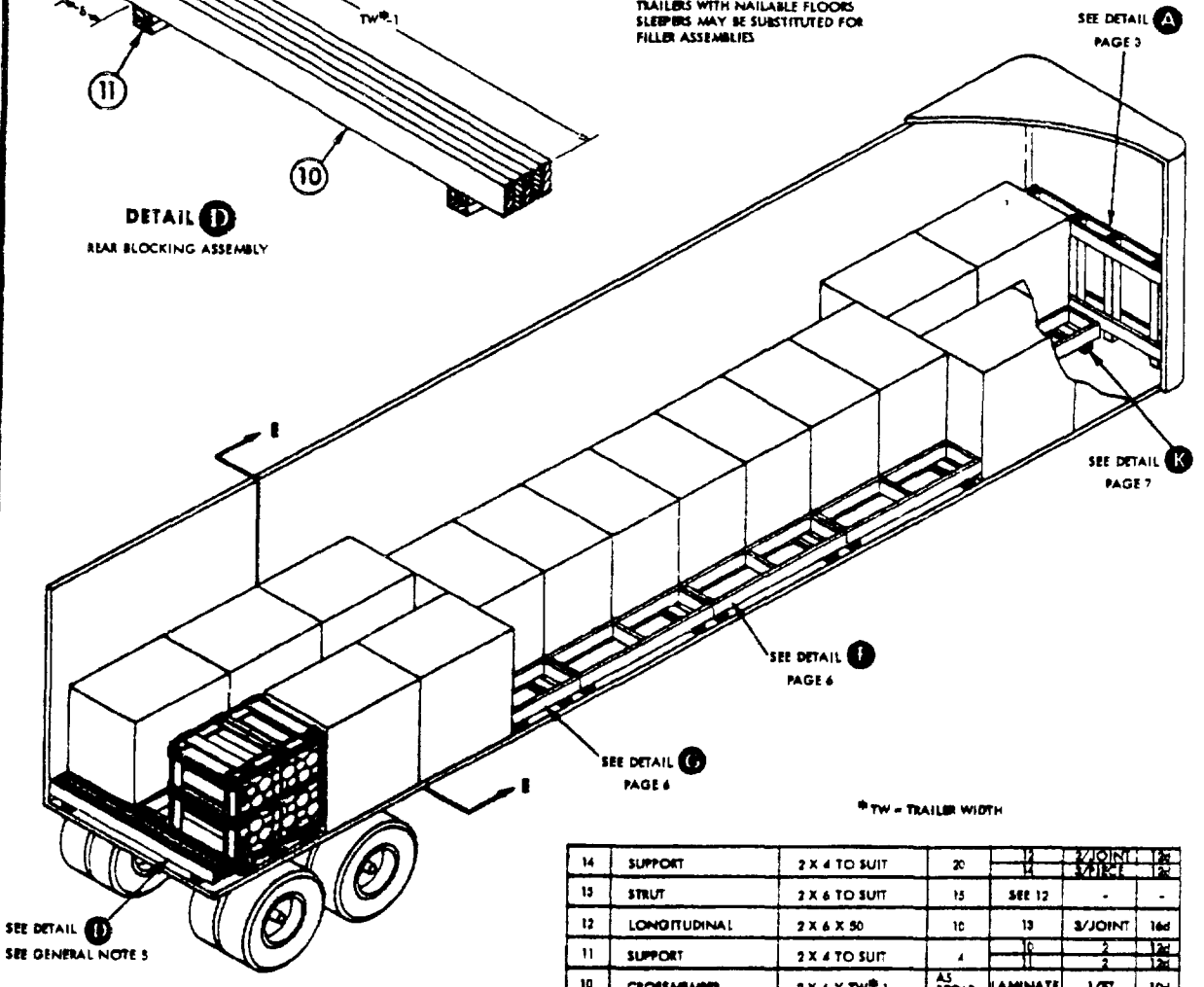
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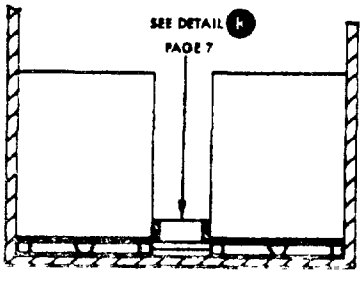
DETAIL D
REAR BLOCKING ASSEMBLY

NOTE

THIS TRUCKLOAD IS FOR TRAILERS WITH NON-NAILABLE FLOORS. FOR TRAILERS WITH NAILABLE FLOORS SLEEPERS MAY BE SUBSTITUTED FOR FILLER ASSEMBLIES



*TW = TRAILER WIDTH

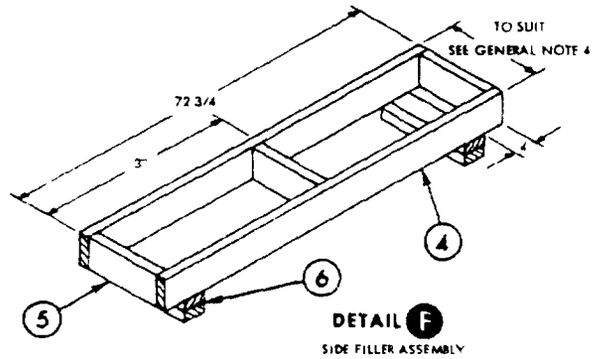


SECTION E-E

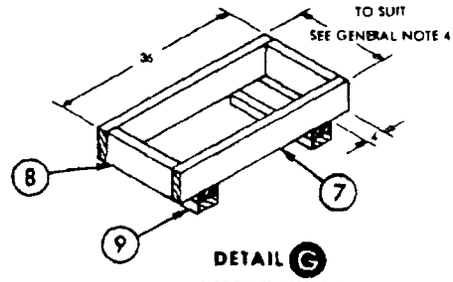
PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
14	SUPPORT	2 X 4 TO SUIT	20	12	2/JOINT	12d
				14	3/STRUT	12d
13	STRUT	2 X 6 TO SUIT	15	SEE 12	-	-
12	LONGITUDINAL	2 X 6 X 50	10	13	3/JOINT	16d
11	SUPPORT	2 X 4 TO SUIT	4	10	2	12d
				11	2	12d
10	CROSSMEMBER	2 X 6 X TW*1	AS REQ'D	LAMINATE	1/FT	10d
9	SUPPORT	2 X 4 TO SUIT	8	7	2/JOINT	12d
				8	3/STRUT	12d
8	STRUT	2 X 6 TO SUIT	4	SEE 7	-	-
7	LONGITUDINAL	2 X 6 X 36	4	8	3/JOINT	16d
6	SUPPORT	2 X 4 TO SUIT	24	4	2/JOINT	12d
				6	3/STRUT	12d
5	STRUT	2 X 6 TO SUIT	18	SEE 4	-	-
4	LONGITUDINAL	2 X 6 X 72 3/4	12	5	3/JOINT	16d
3	AFT CROSSMEMBER	2 X 6 X TW*1	2	2	3/JOINT	12d
2	UPRIGHT	4 X 4 X 90	4	SEE 2 & 3	-	-
1	FWD CROSSMEMBER	2 X 6 TO SUIT	2	2	3/JOINT	12d

LIST OF MATERIALS & NAILING DATA

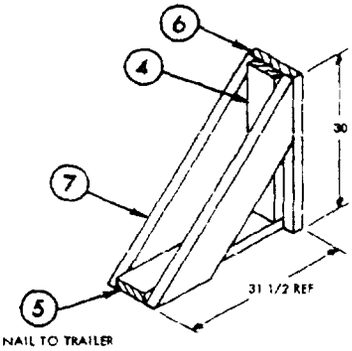
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DETAIL F
SIDE FILLER ASSEMBLY

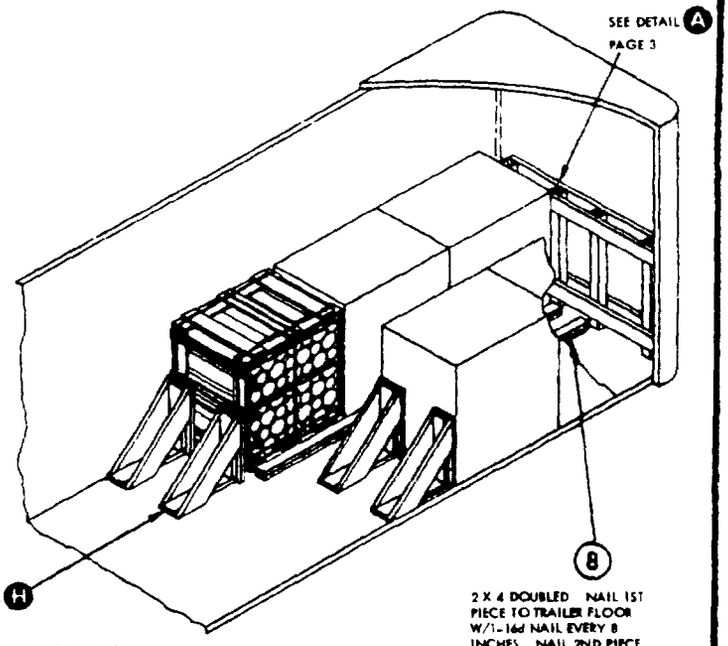


DETAIL G
SIDE FILLER ASSEMBLY



NAIL TO TRAILER FLOOR W/8-16d NAILS

DETAIL H
REAR BRACE ASSEMBLY
(4 REQUIRED)



2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

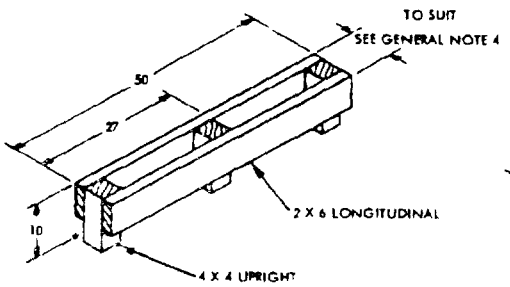
*TW = TRAILER WIDTH

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ D	NAIL TO	NUMBER NAILS	SIZE
8	SLEEPER	2 X 4 X 45	10	SEE FIELD NOTE		
7	BRACE DIAGONAL	2 X 6 TO SUIT	8	4 & 5	3/JOINT	16d
6	BRACE BACKUP	2 X 10 X 30	4	4	5	10d
5	BRACE CLEAT	2 X 6 X 30	4	4	3	16d
4	BRACE UPRIGHT	2 X 6 X 28 1/2	4	SEE 5 & 6		-
3	AFT CROSSMEMBER	2 X 6 X TW*-1	2	2	3/JOINT	12d
2	UPRIGHT	4 X 4 X 50	4	SEE 1 & 3	-	-
1	FWD CROSSMEMBER	2 X 6 X TO SUIT	2	2	3/JOINT	12d

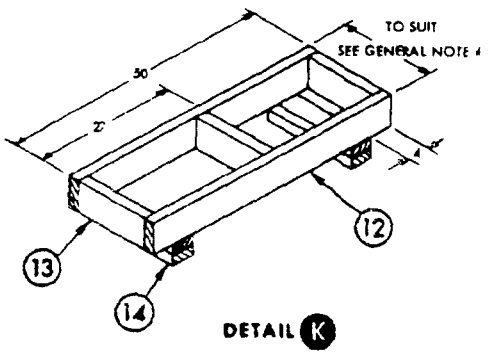
LIST OF MATERIALS & NAILING DATA

TYPICAL LTL (VAN)

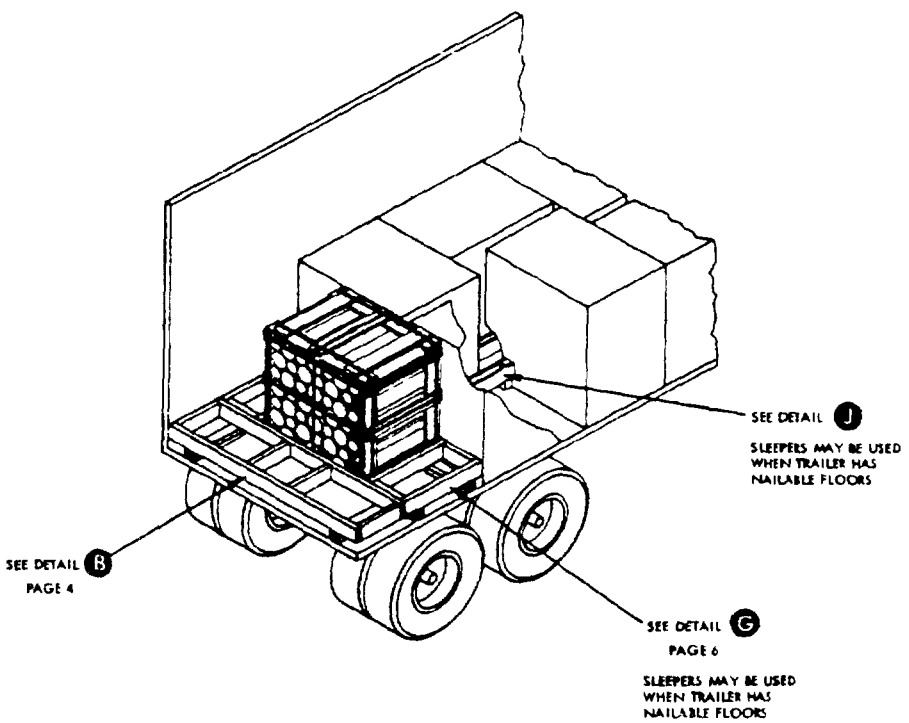
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DETAIL J
CENTER FILLER ASSEMBLY
USED ON PAGE 5



DETAIL K
CENTER FILLER ASSEMBLY
USED ON PAGE 5



REAR BLOCKING PROCEDURE
19 UNIT LOADS IN 40/42 FT TRAILERS

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OMB Approval
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CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

COMMANDING OFFICER
NAVAL WEAPONS STATION EARLE
NAVAL WEAPONS HANDLING LABORATORY
COLTS NECK, NEW JERSEY 07722

FOLD

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